07-18 第2篇

阅读理解（说明文） Why Do Leaves Change Colour in Autumn?

① Every year in late September or October, something magical happens in many parts of the world. Green leaves slowly turn yellow, orange, red or even deep purple. Mountains and parks become a huge, living painting. But have you ever wondered **why** leaves change colour? The answer is a clever mix of sunlight, temperature and chemistry inside every tiny leaf.

② To understand the change, we first need to know what makes leaves green in spring and summer. Inside the leaves there is a green substance called **chlorophyll**. It helps the tree use sunlight, water and air to make food — a process called **photosynthesis**. Because there is so much chlorophyll in a healthy leaf, we only see green and nothing else, even though other colours are already hiding inside.

③ So the other colours are not new. They are there all the time. Yellow and orange come from chemicals called **carotenoids**, the same ones that make carrots orange and bananas yellow. In summer, the strong green of chlorophyll covers these colours completely. It is a little like painting a yellow wall with thick green paint: you see only the green, but the yellow is still there underneath.

④ When autumn arrives, days become shorter and nights become cooler. The tree gets a clear message from nature: winter is coming, and water will soon be hard to get. To save energy, the tree slowly stops making chlorophyll. The green colour begins to fade. As the green gets weaker, the yellow and orange that were hiding for months finally show up. This is why poplar and ginkgo leaves turn bright yellow first.

⑤ Red and purple, however, are a different story. These colours come from **anthocyanins**, which are **not** in the leaf all summer — the tree actually produces them in autumn. Why? Scientists are still studying this question, but one popular idea is that red colours work like a pair of sunglasses for the leaf. They protect the tree from strong autumn sunlight while it quietly moves the last food and sugar from the leaves into the branches, to be saved for next spring. Cool nights and sunny days make more red appear, which is why maple trees look so brilliant after a clear, chilly weekend.

⑥ Finally, the leaves fall. A thin layer slowly grows between each leaf and the branch, cutting the "pipe" that carries water. Without water, the leaf dries up, turns brown, and drops to the ground. Yet this is not the end of the story. Fallen leaves rot into the soil and give food back to the roots. So next spring, the same tree — using the same

sunlight, water and air — will grow new green leaves, and the quiet, beautiful cycle will begin again.

⑦ Next time you walk under a golden or red tree, take a closer look. The colours you see are not a trick of the season. They are the tree's calm, careful preparation for winter — and a quiet promise that spring will come.

根据文章内容，选择最佳答案。

1. Why do leaves look green in spring and summer?

A. Because the sunlight in summer is green.
B. Because leaves have a lot of chlorophyll that covers other colours.
C. Because the tree paints them green every morning.
D. Because there are no other colours inside the leaves at all.

2. Which chemical is responsible for the yellow and orange colours in autumn leaves?

A. Chlorophyll.
B. Water.
C. Carotenoids.
D. Oxygen.

3. According to the passage, why do some leaves turn red or purple in autumn?

A. Because the tree suddenly becomes ill.
B. Because red paint falls from the sky.
C. Because the tree starts to make a new chemical called anthocyanins, possibly to protect the leaf like sunglasses.
D. Because the wind pushes red dust onto them.

4. What happens just before leaves finally fall off the tree?

A. The tree pours hot water on every leaf.
B. A thin layer grows and cuts off the water supply to the leaf.
C. Birds cut the leaves with their beaks.
D. The leaves suddenly become wet and heavy.

5. What is the main purpose of this passage?

A. To sell red and yellow paint for artists.
B. To explain the scientific reasons why leaves change colour in autumn.
C. To tell a sad story about dying trees.
D. To teach readers how to grow maple trees at home.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "Because there is so much chlorophyll..."
【解析】叶绿素含量高，遮盖了其他颜色。A/C 荒谬；D 与原文反（其他颜色存在）。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段 "Yellow and orange come from... carotenoids..."
【解析】类胡萝卜素带来黄橙色。A 为绿色色素；B/D 非色素。故选 C。

3.
【答案】C
【题型】推理判断题
【定位】第⑤段 "anthocyanins... work like... sunglasses..."
【解析】花青素由树秋天产生，可能保护叶片。A/B/D 均无据或荒谬。故选 C。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段 "A thin layer slowly grows... cutting the 'pipe'..."
【解析】一层薄膜切断水分通道。A/C/D 均无据。故选 B。

5.
【答案】B
【题型】主旨大意题
【定位】通篇
【解析】全文讲解树叶秋天变色的科学原因。A/C/D 均非主旨。故选 B。

词汇注释（超纲词）

- **chlorophyll** [n.] 叶绿素
- **photosynthesis** [n.] 光合作用
- **carotenoid** [n.] 类胡萝卜素
- **anthocyanin** [n.] 花青素
- **poplar / ginkgo / maple** [n.] 杨树 / 银杏 / 枫树
- **rot** [v.] 腐烂